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Appendix

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A FURTHER EVALUATION OF SYNTHETIC-DATA INTERVENTION

A.1 SYNTHETIC-DATA INTERVENTION DOES NOT AFFECT PERFORMANCE ON BENCHMARKS

As shown in Appendix A.5, synthetic-data intervention is most-effective when a small amount of instruction-tuning data is included with our generated data during finetuning. For this reason, we expect that models should not forget prior learned information and should retain their abilities in benchmark settings that were achieved via instruction tuning. We show this by examining model performance on the MMLU (Hendrycks et al., 2021) and BIG-Bench Hard (Suzgun et al., 2022) benchmarks in a 5-shot and 3-shot setting, respectively, following Chung et al. (2022).

In Figure 7, we show model performance on these two benchmarks before and after intervention. We see that synthetic-data intervention results in a performance change of -1.6% (Flan-cont-PaLM-62B on MMLU) to $+0.6\%$ (Flan-PaLM-540B on BIG-Bench Hard). We found, however, that continuing the instruction-tuning procedure (i.e., 100% of tuning data is instruction-tuning data) for another 1k steps can lead to performance changes of -3.6% (Flan-cont-PaLM-62B on MMLU) to $+0.7\%$ (Flan-PaLM-8B on MMLU). For this reason, we conclude that the performance change from intervention does not indicate any actual difference in abilities, which is an expected result because we mixed in instruction-tuning data as part of our finetuning procedure.

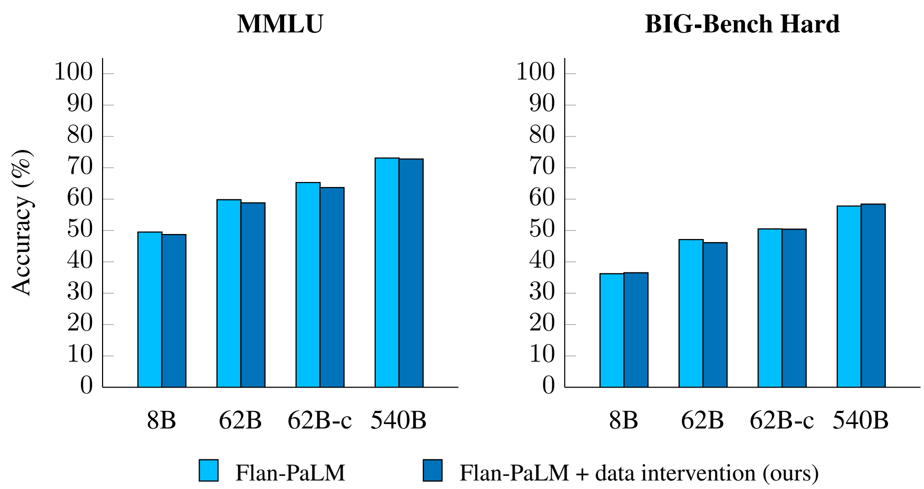

Figure 7: Performance on MMLU and BIG-Bench Hard does not significantly change after synthetic-data intervention. Accuracy shown is an unweighted average over all tasks for each benchmark (per-task results are shown in Appendix D.1 and Appendix D.2).

A.2 SYNTHETIC-DATA INTERVENTION DOES NOT AFFECT CHAIN-OF-THOUGHT REASONING

One limitation of our synthetic-data intervention is that it does not include any data that uses chain-of-thought reasoning (Wei et al., 2022b, CoT) because sycophancy tasks are set in a zero-shot setting. We thus aim to ensure that our method does not result in any performance loss in CoT settings. To analyze this, we reformat prompts from the two benchmarks in Appendix A.1 to include CoT prompting, and we then compare model performance before and after applying intervention. We used the same CoT prompts as Chung et al. (2022).

These results are shown in Figure 8. Overall, we see that there is no significant increase or decrease in performance—synthetic-data intervention results in performance changes of between -1.5% (Flan-cont-PaLM-62B on MMLU) to $+3.1\%$ (Flan-PaLM-8B on MMLU). While the maximum performance improvement seems large, we stress that a definitive conclusion of improvement cannot be drawn because continued instruction tuning for 1k steps results in performance differences of up to -4.7% (Flan-cont-PaLM-62B on MMLU). At the same time, the findings seem to indicate that, at the minimum, there was no loss in CoT abilities due to intervention.